1.3.3 Enabling automation

Because AWS is API driven, you can automate everything: write code to create networks, start virtual machine clusters, or deploy a relational database. Automation increases reliability and improves efficiency.

The more dependencies your system has, the more complex it gets. A human can quickly lose perspective, whereas a computer can cope with interconnected systems of any size. You should concentrate on tasks humans are good at—such as describing a system—while the computer figures out how to resolve all those dependencies to create the system. Setting up an environment in the cloud based on your blueprints can be automated with the help of infrastructure as code, covered in chapter 4.

1.3.4 Flexible capacity (scalability)

Flexible capacity reduces overcapacity. You can scale from one virtual machine to thousands of virtual machines. Your storage can grow from gigabytes to petabytes. You no longer need to predict your future capacity needs for the coming months and years to purchase hardware.

If you run a web shop, you have seasonal traffic patterns, as shown in figure 1.7. Think about day versus night, and weekday versus weekend or holiday. Wouldn’t it be nice if you could add capacity when traffic grows and remove capacity when traffic shrinks? That’s exactly what flexible capacity is about. You can start new virtual machines within minutes and throw them away a few hours after that.